## Important Array Interview Questions :→

### 1. Prefix Sum :→

arr = {1, 2, 3, 4};
0 1 2 3

Definition : The sum till the current index in an array is called Prefix Sum of that array :

Output :→ prefix = {1, 3, 6, 10};

prefix = {0, 0, 0, 0};
= {1, 3, 6, 10};
0 1 2 3

```
prefix[0] = arr[0];
for (int i = 1; i < prefix.length; i++) {
    prefix[i] = prefix[i-1] + arr[i];
}
```

P[3] = P[2] + a[3]

O(n) int[] arr = {1, 2, 9, 6, -1, -2, 8, 7, 4}
0 1 2 3 4 5 6 7 8

O(n) mx = arr[0] = 1
mn = arr[0] = 1

mx = ~~1~~ 9

for for

```
for (1 to last) {
    if (arr[i] > mx)
        mx = arr[i]
    if (arr[i] < mn)
        mn = arr[i]
}
print (mx, mn)
```

O(n²)

n elements

Time Complexity $\boxed{O(n)}$

One for loop Till n elements

mn = ~~1~~
~~-1~~
-2

-1 < 1
-2 < -1

### Difference array :→ Add a particular number to some elements in a given range inside the array.

int val = 10 {Range 1 to 3}

int n = 5
int[] diff = new int[5];

Step 1: diff[1] += val;
0 + = 10
= 10;

{0, 0, 0, 0, 0} {0, 0, 0, 0, 0};
0 1 2 3 4
0 1 2 3 4

{0, 10, 0, 0, 0}
0, 10, 10, 10, 10

O/P {0, 10, 10, 10, 0};

int l = 1
int r = 3
if (r+1 < n) (1+3) 4 < 5
diff[r+1] -= val;

{1, 4, 5, 8, 6}
+10 (1-3) {1, 14, 15, 18, 6}
(l) 1 ⟷ 3 (r)

[l = 1, r = 3;]
int val = 10;
[if (r+1) < n]
3+1 = 4 < 5
diff[r+1] -= val;
diff[4] -= 10
0 - 10 = -10

diff {0, 10, 0, 0, -10}; n = 5
0 1 2 3 4
diff[l] += val;
diff[1] += val;
0 + 10 = 10
diff {0, 10, 0, 0, -10}

add a negative val to eliminate the addition.

Apply prefix sum:
{0, 10, 10, 10, 0}

{target = 9}

arr = {1, 3, 2, 7}
0 1 2 3

```
for (int i = 0; i < n; i++)
    for (int j = i+1; j < n; j++)
        if (arr[i] + arr[j] == target;
        print (i & j)
        print a[i] & a[j]
```

i = 0
a[0] = 1
j = 0+1 = 1 (3)

→ 2 & 3

2 7

1+3

DRYRUN

### Majority Element : {Boyer Moore's Voting Algorithm}